# VintageBench: Do AI Analysts Update Correctly When the Data Are Revised?

Divya Jyoti, MIMS (AI & Data Science), UC Berkeley · github.com/divyajyoti-dev/vintagebench-release · 6 months, Nov 2026 – Apr 2027

## Problem

Analysts and AI agents draw conclusions from provisional data such as first-release GDP, payrolls and industrial production, which statistical agencies later revise. A trustworthy analyst agent must answer from the data available on the date in question, judge how fragile its conclusion is, and update when, and only when, revisions change it.

## Task format and grading

- Plain yes/no questions with an as-of date, e.g. *"Did US real GDP shrink in 2022 Q2?"*. In turn 1 a **tool serves only the vintage published by that date**. The reply gives the calculation, the answer, the confidence, and the **probability the conclusion survives full revision**.
- Turn 2 is neutral: *"Here is the latest data. Answer the question."* The tasks are flips (the conclusion changes on revision), controls, and no-change tasks (the turn-2 data are identical).
- Gold answers are computed from the vintages by code, with no human labels and no LLM judge. Metrics: as-of and revised accuracy; update correctness (actual answer change); survival calibration and discrimination; leakage; label/text contradiction; and a no-tools memorization probe. Sanity agents validate the grader.
- Data: Philadelphia Fed Real-Time Data Set for Macroeconomists (BLS, BEA and Federal Reserve Board series). It is fetched by a checksummed script, so no raw data are redistributed. A v0 prototype used ALFRED.

## Pilot (v1): 45 tasks × 4 OpenAI configs × 3 trials = 540 episodes, $2.83

**Base rate.** Of 1,235 candidate questions, 120 (9.7%) first-release conclusions differ on the latest vintage. For the payroll 3-month-average >150k template it is 23/116 (19.8%).

**Calibration.** Mean survival probabilities were 0.86–0.92, close to the ~90% survival the base rate implies. But they barely separate conclusions that later flipped from those that held (AUC flip vs non-flip, with ties counting half):

| Config | Mean survival prob., flips | Mean, non-flips | AUC | Flips rated ≥ 0.8 | As-of correct | Update correct (flips) |
|---|---|---|---|---|---|---|
| gpt-6-sol / reasoning off | 0.85 | 0.88 | 0.61 | 48/60 | 135/135 | 60/60 |
| gpt-6-sol / medium | 0.85 | 0.89 | 0.58 | 45/60 | 135/135 | 60/60 |
| gpt-6-luna / reasoning off | 0.86 | 0.95 | 0.64 | 53/60 | 109/135 | 41/60 |
| gpt-6-luna / medium | 0.91 | 0.93 | 0.60 | 54/59 | 133/135 | 59/60 |

**Memorization (no tools).** Recalls that matched a vintage matched the first release far more often than the latest: 71 vs 17 of 324 distinguishable values. Many matches are on low-precision quantities, where they can be chance. With tools, leakage was 0/540.

## Lessons

- v0 showed that frontier models ace questions that spell out the formula. On v1's plain questions the frontier model is still at 135/135. The small model with reasoning off failed mainly at data gathering (20 of 26 wrong answers). Accuracy alone therefore misses the hard part: **judgement about revisions**.
- With the calculation placed before the answer, label/text contradictions dropped to 0 of 1,080 turns. In v0 they were the main error source.
- Limits: small n, a thin flip pool, only OpenAI models, and survival scored against the latest pinned vintage.

## Expected outputs and impact

An open benchmark of ~600 tasks over more series, revision horizons and multi-step analyses; the tool-serving harness; base-rate tables; leaderboard results for frontier and open-weight models; and a paper. Agents used in economic, financial and policy analysis need to know when their conclusions are provisional. The same design generalizes to **earnings restatements, corrected clinical data and data-pipeline backfills**.

## How it differs

*Total Recall?* (Crane, Karra & Soto, FRB FEDS 2025-044, June 2025) shows that LLM recall mixes first-print data with later revisions, and that LLMs often believe they have data not yet released. It probes memory. VintageBench tests agents working from dated, tool-served vintages, and whether they update correctly. ExAnte (arXiv:2505.19533) and Look-Ahead-Bench (arXiv:2601.13770) measure temporal leakage. LongDS-Bench (arXiv:2605.30434) and TrustDABench (arXiv:2608.24145) test analytical state and robustness. None uses real data vintages.

## Methodology and request

- **Expert data:** economists and statistical-agency analysts review the templates and gold conventions, write harder multi-step questions, and rate which revisions an expert would anticipate, giving a human survival baseline.
- **Snorkel support:** engineering for task generation at scale, hosting, a leaderboard, and restatement and clinical-data harness extensions.
- **Credits: $2,000.** The costliest pilot configuration cost $1.52 for 135 episodes ($0.0113 each). 600 tasks × 3 trials × 10 configs ≈ $203 per pass. Five passes come to $1,017, doubled for longer agentic tasks.

## Milestones

| Nov 2026 | Dec 2026 – Jan 2027 | Feb 2027 | Mar 2027 | Apr 2027 |
|---|---|---|---|---|
| Expert review; more series | 600 tasks; open-weight models | Restatement and clinical pilots | Human baseline; leaderboard | Paper; public release |

## Team

Divya Jyoti (lead): MIMS (AI & Data Science), UC Berkeley. About 3 years of software engineering in quantitative finance at UBS. Built an evaluation framework for a text-to-SQL data-analysis agent during an industry internship. Open to a Snorkel co-advisor.

Data: Federal Reserve Bank of Philadelphia, Real-Time Data Set for Macroeconomists (Croushore & Stark, J. Econometrics 2001). The Bank and the source agencies do not endorse this work.